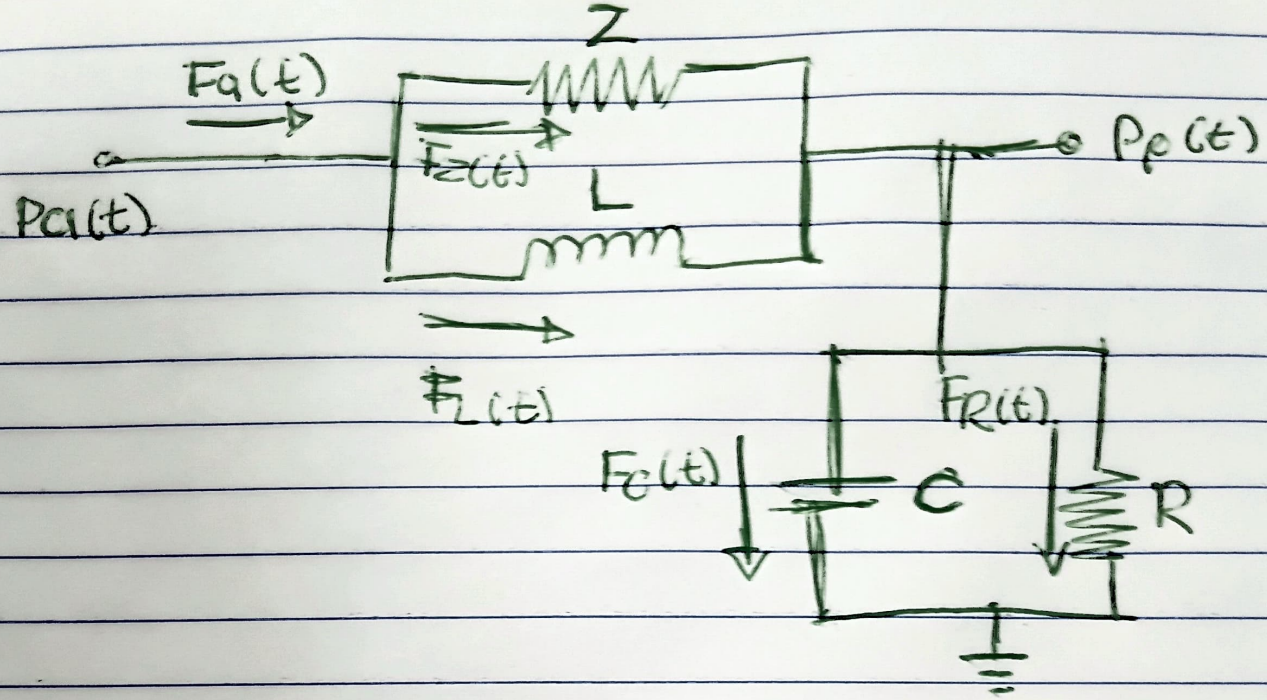

Z
Fa(t)
Pa(t)
Fz(t)
L
Pp(t)
FL(t)
FR(t)
Fc(t)
C
R

Función de transferencia

Ecuaciones principales

$$F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_Z(t) = (P_a(t) - P_p(t))/Z$$
$$F_L(t) = \frac{1}{L}\int (P_a(t) - P_p(t))\,dt$$
$$F_C(t) = C\, d(P_p(t))/dt$$
$$F_R(t) = P_p(t)/R$$

Transformada de Laplace

$$F_a(s) = F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$F_Z(s) = (P_a(s) - P_p(s))/Z$$
$$F_L(s) = (P_a(s) - P_p(s))/Ls$$
$$F_C(s) = Cs\,P_p(s)$$
$$F_R(s) = P_p(s)/R$$

Procedimiento algebraico

$$F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} + \frac{P_a(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R} + \frac{P_p(s)}{Z} + \frac{P_p(s)}{Ls}$$

$$P_a(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = P_p(s)\left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls}\right)$$

$$P_a(s)\left(\frac{Ls + Z}{ZLs}\right) = P_p(s)\left(\frac{Cs^2RZL + ZLs + RLs + RZ}{RZLs}\right)$$

$$\frac{Pb(s)}{Pa(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CRZLs^2 + L(Z+R)s + RZ}{RZLs}}$$

$R(Ls+Z)$

Resultado

$$\frac{Pb(s)}{Pa(s)} = \frac{RLs + RZ}{CRZLs^2 + L(Z+R)s + RZ}$$

Estabilidad del sistema en lazo abierto

$$CRZLs^2 + L(Z+R)s + RZ$$

$a = CRZL \quad b = L(Z+R) \quad c = RZ$

Hipotenso

$Z = 0.010$
$C = 3.500$
$R = 1.000$
$L = 0.001$

$\lambda_1 = -14.42857 + 8.80514i$
$\lambda_2 = -14.42857 - 8.80514i$

El sistema presenta una respuesta estable y subamortiguada a un escalón unitario de entrada

Normotenso

$Z = 0.010$
$C = 1.500$
$R = 1.000$
$L = 0.001$

$\lambda_1 = -12.0616$
$\lambda_2 = -55.2717$

El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada

Hipertenso

$Z = 0.010$
$C = 1.000$
$R = 1.000$
$L = 0.001$

$\lambda_1 = -11.1267 \quad \lambda_2 = -89.8732$

El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada

Error en estado estacionario

$$e(s) = \lim_{s \to 0} sPa(s)\left(1 - \frac{Pp(s)}{Pa(s)}\right)$$

$$e(s) = \lim_{s \to 0} s\frac{1}{s}\left(1 - \frac{RLs + RZ}{CRZLs^2 + L(Z+R)s + RZ}\right) = \emptyset$$

$$e(s) = 0\ V$$

Modelo de ecuaciones integro-diferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{Pa(t) - Pp(t)}{Z} + \frac{1}{L}\int (Pa(t) - Pp(t))\,dt = C\frac{dPp(t)}{dt} + \frac{Pp(t)}{R}$$

$$\frac{Pa(t)}{Z} - \frac{Pp(t)}{Z} + \frac{1}{L}\int (Pa(t) - Pp(t))\,dt = C\frac{dPp(t)}{dt} + \frac{Pp(t)}{R}$$

$$\frac{Pp(t)}{R} + \frac{Pp(t)}{Z} = \frac{Pa(t)}{Z} + \frac{1}{L}\int (Pa(t) - Pp(t))\,dt - C\frac{dPp(t)}{dt}$$

$$Pp(t)\left(\frac{Z+R}{RZ}\right) = \frac{Pa(t)}{Z} + \frac{1}{L}\int (Pa(t) - Pp(t))\,dt - C\frac{dPp(t)}{dt}$$

$$Pp(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int (Pa(t) - Pp(t))\,dt - C\frac{dPp(t)}{dt}\right]\frac{RZ}{Z+R}$$